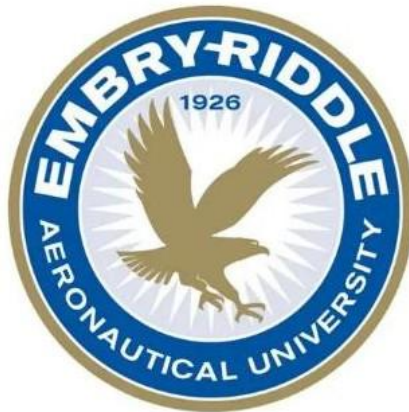


Speed Control



Professor: Bernardo Restrepo

Graduate Assistant: Jared Getz

Author: Anuranan Bharadwaj

AE 443: Experimental Dynamics and Control Laboratory

Department of Aerospace Engineering

Embry-Riddle Aeronautical University

Daytona Beach, FL, 32114

April 17, 2026

Procedure

A. Step Response with PI Control

Main Goal:

The goal of this experiment is to design and evaluate a PI controller for regulating the speed of the SRV02 motor. The objective is to achieve accurate tracking of the reference speed while minimizing steady-state error and controlling the transient response.

Procedure:

In simulation, the PI gains k_p and k_i are applied in the Simulink model of the motor. A square-wave reference input is used to generate step changes in speed, and the response is recorded to determine peak time, percent overshoot, and steady-state error.

In implementation, the same controller is applied to the physical SRV02 system using the QUARC interface. The motor speed is measured in real time, and the response is recorded to compute the same performance metrics and compare with the simulated results.

B. Step Response with Lead Control

Main Goal:

The goal of this experiment is to evaluate the performance of a lead compensator as an alternative to PI control, with a focus on improving transient response and system stability.

Procedure:

In the simulation, the lead controller is implemented in the Simulink model using the designed parameters. The same square-wave reference input is applied, and the speed response is recorded to determine peak time, percent overshoot, and steady-state error.

In implementation, the lead controller is applied to the SRV02 hardware under the same operating conditions. The measured speed response is recorded and used to calculate performance metrics, allowing comparison between simulated and experimental behavior.

Results

A. Step Response with PI Control

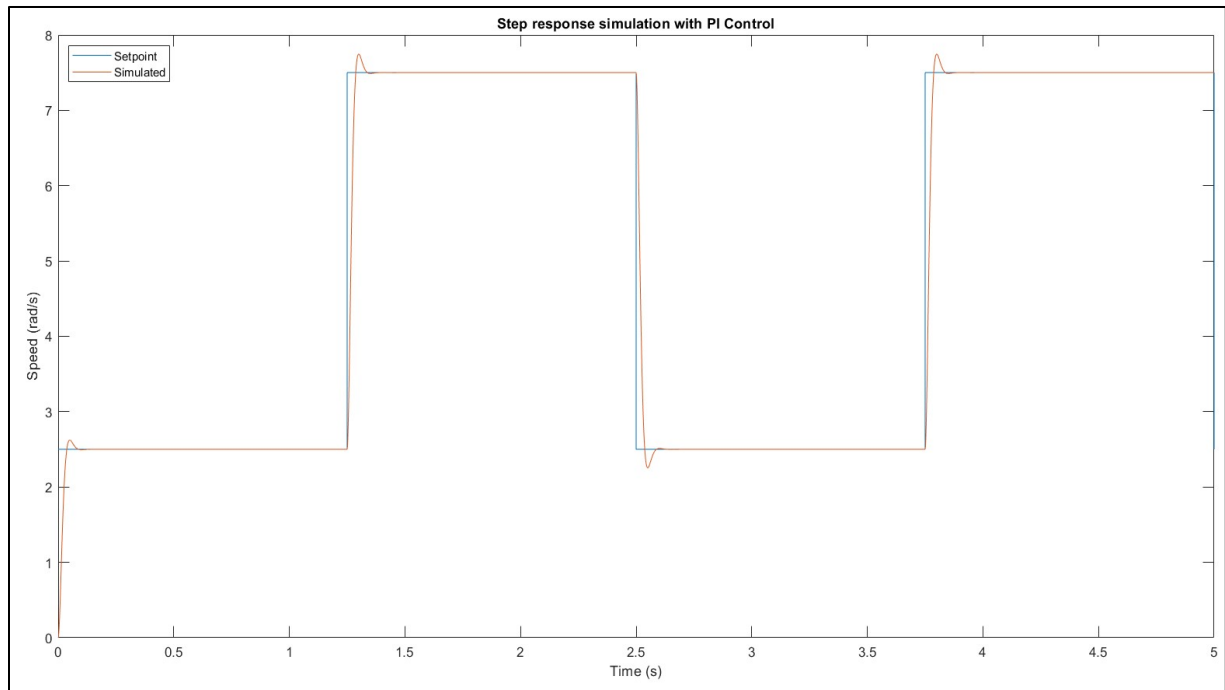


Figure 1: Step response simulation with PI Control – Speed vs Time

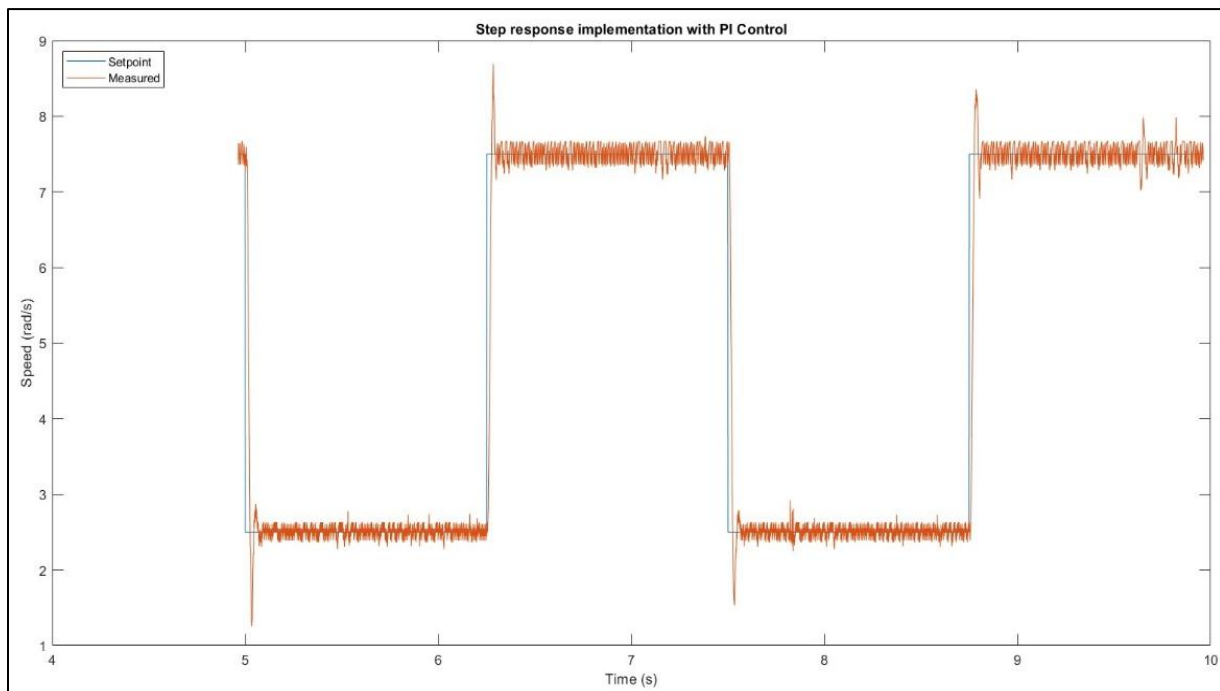


Figure 2: Step response implementation with PI Control – Speed vs Time

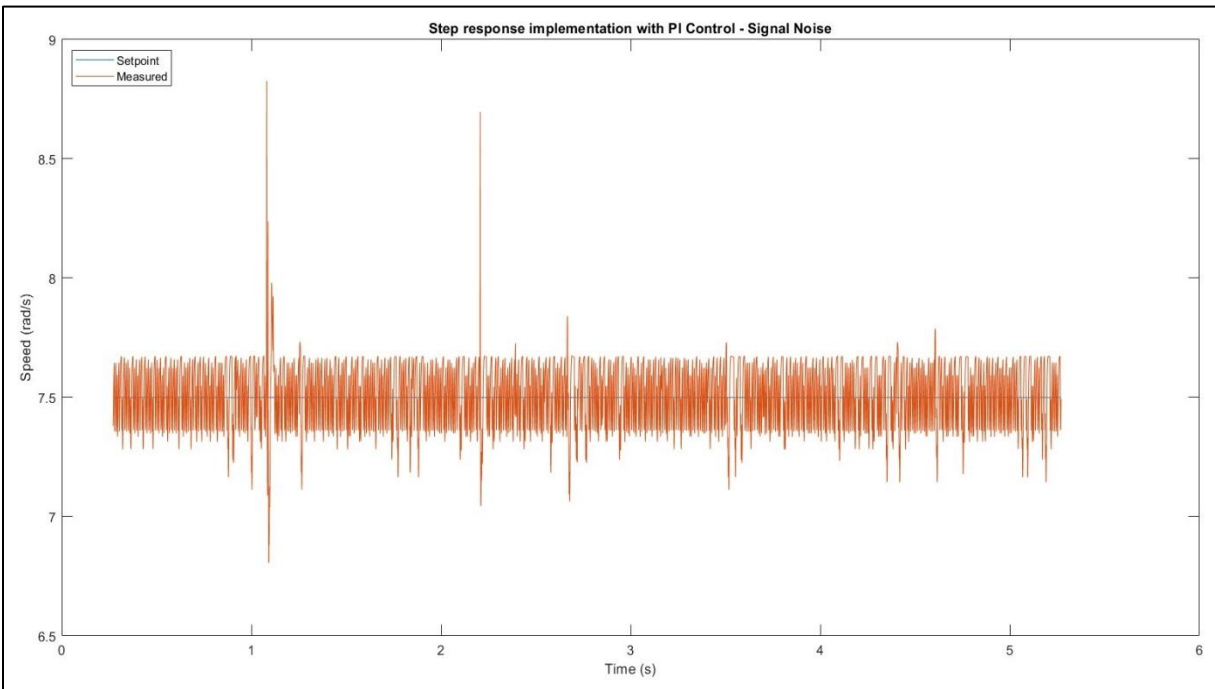


Figure 3: Step response implementation with PI Control – Signal Noise

B. Step Response with Lead Control

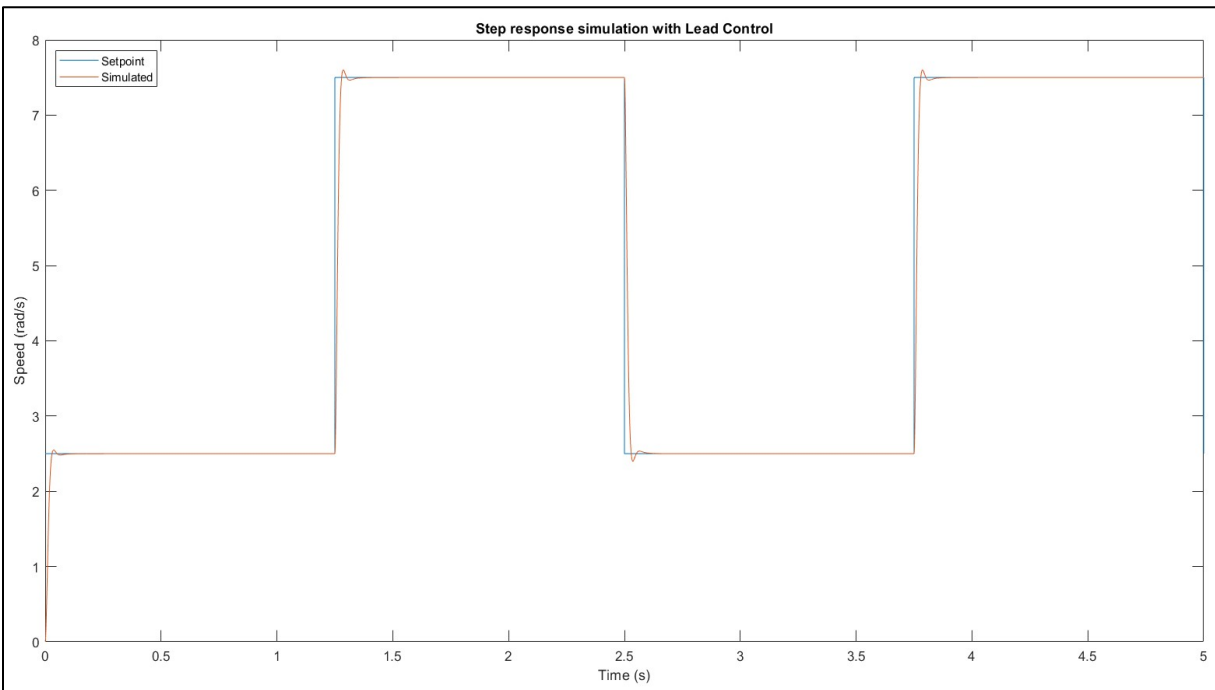


Figure 4: Step response simulation with Lead Control – Speed vs Time

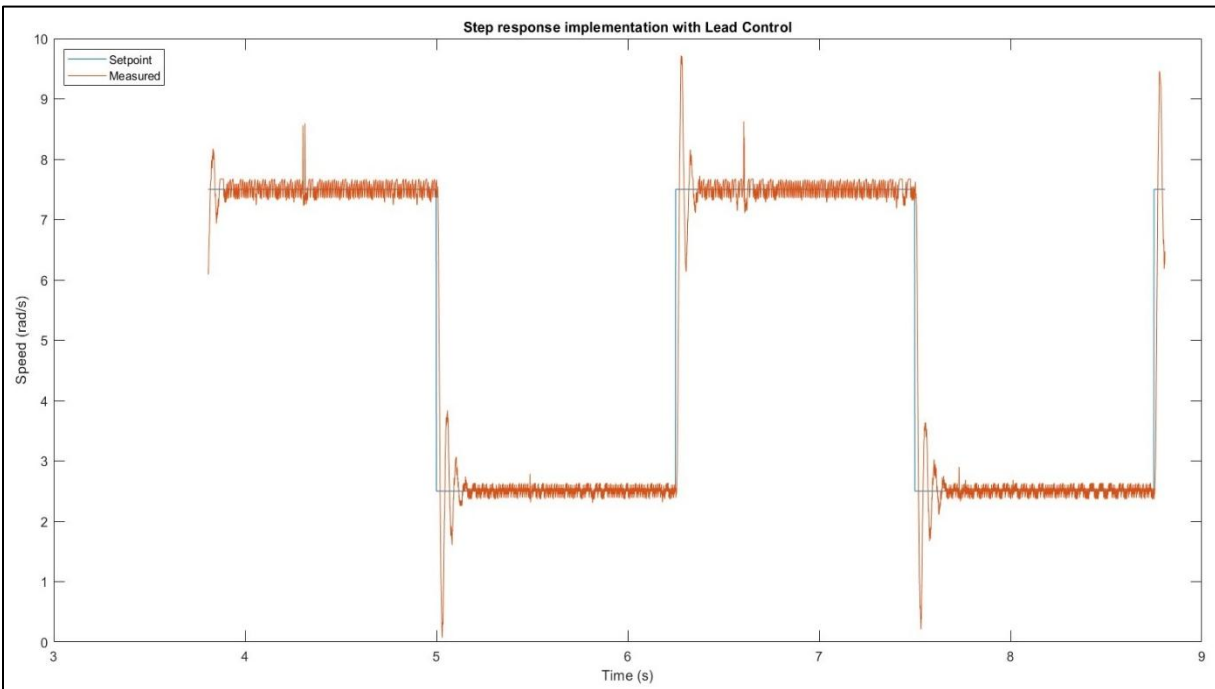


Figure 5: Step response implementation with Lead Control – Speed vs Time

Table 1: Summary of Results

Description	Symbol	Value	Unit
PI Step Response Simulation			
Peak Time	t_p	0.04	s
Percent Overshoot	PO	4.4	
Steady-State Error	e_{ss}	0	rad/s
PI Speed Control Implementation			
Measured peak-to-peak ripple	$e_{w,meas}$	1.16	rad/s
Steady-State Error	e_{ss}	-0.0000317	rad/s
Peak Time	t_p	0.034	s
Percent Overshoot	PO	23.8	
Step Response Simulation with Lead Control			
Peak Time	t_p	0.04	s
Percent Overshoot	PO	2	
Steady-State Error	e_{ss}	0	rad/s
Lead Speed Control Implementation			

Peak Time	t_p	0.03	s
Percent Overshoot	PO	44.4	
Steady-State Error	e_{ss}	-0.0012	rad/s

Analysis

A. Derivation - Frequency response magnitude

The objective is to determine the transfer function $P_i(s)$, its frequency response, and assess stability using gain and phase margins.

The SRV02 motor speed dynamics are modeled as a first-order system:

$$\tau \dot{\omega}_l(t) + \omega_l(t) = K V_m(t) \quad (1)$$

Taking the Laplace transform (zero initial conditions),

$$(\tau s + 1)\Omega_l(s) = K V_m(s) \quad (2)$$

which gives the plant transfer function

$$P(s) = \frac{\Omega_l(s)}{V_m(s)} = \frac{K}{\tau s + 1} \quad (3)$$

In this experiment, the transfer function of interest includes an integrator:

$$P_i(s) = \frac{P(s)}{s} \quad (4)$$

Substituting the plant model,

$$P_i(s) = \frac{K}{s(\tau s + 1)} \quad (5)$$

Using the nominal parameters for the system, $K = 1.53$ and $\tau = 0.0254$, the transfer function becomes

$$P_i(s) = \frac{1.53}{s(0.0254s + 1)} \quad (6)$$

To evaluate the frequency response, substitute $s = j\omega$:

$$P_i(j\omega) = \frac{K}{j\omega(1 + j\tau\omega)} \quad (7)$$

The magnitude is computed as

$$|P_i(j\omega)| = \frac{K}{\omega \cdot \sqrt{1 + (\tau\omega)^2}} \quad (8)$$

Substituting the values,

$$|P_i(j\omega)| = \frac{1.53}{\omega \cdot \sqrt{1 + (0.0254\omega)^2}} \quad (9)$$

The following Bode plot was generated using the above values:

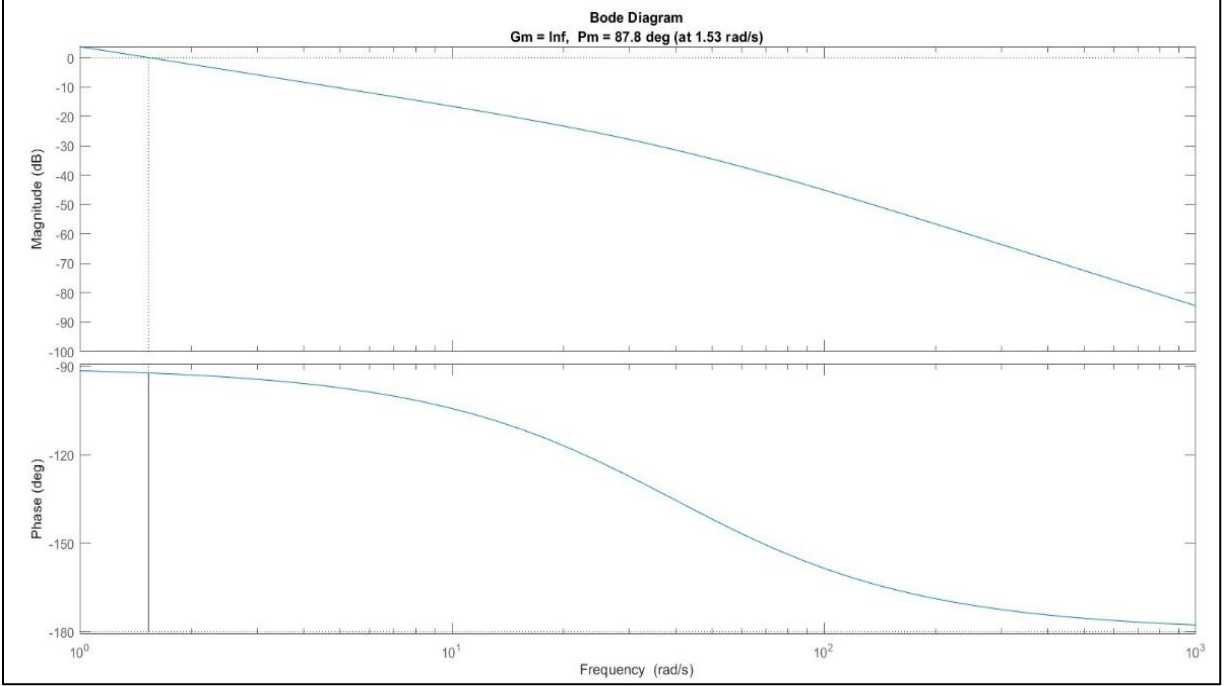


Figure 6: Bode Plot

From the plot, Gain Margin is infinite and Phase Margin is 87.8°. A large positive phase margin indicates that the system can tolerate significant phase lag before becoming unstable. The infinite gain margin occurs because the phase does not reach -180° in a way that defines a finite crossover.

The characteristic equation of $P_i(s)$ is

$$s(\tau s + 1) = 0 \quad (10)$$

which gives poles at

$$s = 0, \quad (11)$$

$$s = -\frac{1}{\tau} = -39.37 \quad (12)$$

Since one pole is exactly at the origin, the system is not asymptotically stable. Instead, the system is marginally stable.

B. Step response simulation with PI Control

The step response of the system with PI control was analyzed using simulation to evaluate how well the controller regulates the motor speed. The response was obtained by applying a square-wave reference signal, where the speed transitions between two setpoints, allowing observation of the transient behavior.

The key performance metrics were determined directly from the simulated speed response plot.

The peak time is calculated using

$$t_p = t_{max} - t_0 \quad (13)$$

From figure 1, the following points were selected, and the peak time was calculated:

$$t_p = 1.29 - 1.25 = 0.04 \text{ s} \quad (14)$$

The percent overshoot is computed using

$$PO = \frac{100(y_{max} - y_f)}{y_f - y_i} \quad (15)$$

Using the values obtained from the plot,

$$PO = \frac{100(7.72 - 7.5)}{7.5 - 2.5} = 4.4 \% \quad (16)$$

This shows that the response slightly exceeds the desired value but remains within acceptable limits.

The steady-state error is determined from the difference between the reference input and the output once the system has settled,

$$e_{ss} = r_{ss} - y_{ss} \quad (17)$$

From figure 1,

$$e_{ss} = 0 \text{ rad/s} \quad (18)$$

which confirms that the output matches the reference exactly at steady state.

C. Step response implementation with PI Control

The step response of the PI-controlled system was obtained using the experimental SRV02 setup. The measured speed data were used to calculate the peak time, percent overshoot, steady-state error, and peak-to-peak ripple.

The peak time is determined using

$$t_p = t_{max} - t_0 \quad (19)$$

From figure 2,

$$t_p = 6.284 - 6.25 = 0.034 \text{ s} \quad (20)$$

The percent overshoot is calculated using

$$PO = \frac{100(y_{max} - y_f)}{y_f - y_i} \quad (21)$$

Using the values obtained from the measured response,

$$PO = \frac{100(8.69 - 7.5)}{7.5 - 2.5} = 23.8 \% \quad (22)$$

The steady-state error is computed as

$$e_{ss} = r_{ss} - y_{ss} \quad (23)$$

Using the average steady-state values,

$$e_{ss} = 4.96861077 - 4.96864250 = -0.0000317 \text{ rad/s} \quad (24)$$

This value was calculated over a time period from 4.9 to 9.9 seconds.

The peak-to-peak ripple in the measured speed signal is calculated using

$$e_{\omega, \text{meas}} = \omega_{max} - \omega_{min} \quad (25)$$

From figure 3,

$$e_{\omega, \text{meas}} = 7.97 - 6.81 = 1.16 \text{ rad/s} \quad (26)$$

The theoretical ripple estimate is calculated using

$$e_{\omega} = \frac{K_n \omega_l}{100} \quad (27)$$

Using $K_n = 7\%$ and $\omega_l = 7.5 \text{ rad/s}$,

$$e_{\omega} = \frac{7 \times 7.5}{100} = 0.525 \text{ rad/s} \quad (28)$$

Comparing the two values,

$$1.16 \text{ rad/s} > 0.525 \text{ rad/s}$$

Thus, the measured ripple is significantly larger than the theoretical estimate.

D. Step response simulation with Lead Control

The simulated step response of the SRV02 with the lead compensator was analyzed using the same time-domain performance measures used for the PI controller, such as peak time, percent overshoot, and steady-state error.

The peak time is found using

$$t_p = t_{max} - t_0 \quad (29)$$

Using figure 4 values,

$$t_p = 1.28 - 1.24 = 0.04 \text{ s} \quad (30)$$

which indicates a fast transient response.

The percent overshoot is calculated using

$$PO = \frac{100(y_{max} - y_f)}{y_f - y_i} \quad (31)$$

From the simulated data,

$$PO = \frac{100(7.6 - 7.5)}{7.5 - 2.5} = 2 \% \quad (32)$$

showing that the peak of the response remains close to the desired final value.

The steady-state error is determined from

$$e_{ss} = r_{ss} - y_{ss} \quad (33)$$

For the simulation, the output reaches the same value as the reference after the response settles, so the steady-state is zero.

E. Step response implementation with Lead Control

The step response of the SRV02 system with the lead compensator was obtained experimentally. The measured speed response was used to determine the peak time, percent overshoot, and steady-state error.

The peak time is calculated using

$$t_p = t_{max} - t_0 \quad (34)$$

where t_0 is the time at which the step input is applied and t_{max} is the time at which the maximum response occurs. From figure 5,

$$t_p = 6.28 - 6.25 = 0.03 \text{ s}$$

which indicates a very fast response of the system to the change in reference speed.

The percent overshoot is determined using

$$PO = \frac{100(y_{max} - y_f)}{y_f - y_i} \quad (35)$$

where y_{max} is the maximum value of the response and y_f is the final steady-state value. From the measured response,

$$PO = \frac{100(9.72 - 7.5)}{7.5 - 2.5} = 44.4 \% \quad (36)$$

This shows that the system response exceeds the desired value significantly before settling.

The steady-state error is calculated using

$$e_{ss} = r_{ss} - y_{ss} \quad (37)$$

where r_{ss} is the reference input and y_{ss} is the measured steady-state output. Using the averaged steady-state values,

$$e_{ss} = 4.9965 - 4.9977 = -0.0012 \text{ rad/s} \quad (38)$$

This value was calculated over a time period from 3.8 to 8.8 seconds. The small magnitude of this value indicates that the output closely matches the reference once the system has settled.

Conclusion

The primary difference between the PI and lead controllers, based on both modeling and implementation findings, is how they maintain a balance between response speed, overshoot, and steady-state accuracy.

From the simulation results, both controllers performed effectively and satisfied the design requirements. The PI controller produced a peak time of 0.04 s, a percent overshoot of 4.4%, and zero steady-state error. The lead controller gave a faster response, with a peak time of 0.04 s, a lower overshoot of 2.0%, and a zero steady-state error. This shows that under ideal

conditions, the lead controller provided a faster and better-damped response than the PI controller.

However, both controllers did not behave as well in the hardware implementation as they did in the simulation. The PI controller had a peak time of 0.034 s, overshoot of 23.8%, and steady-state error of -0.0000317 rad/s. The lead controller responded even faster, with a peak time of 0.03 s, but its overshoot increased to 44.4%, while its steady-state error remained small at -0.0012 rad/s. As a result, even though the lead controller was implemented more quickly than the PI controller, it resulted in a significantly greater overshoot.

Another significant difference is the measured ripple in the real system. For the PI implementation, the measured peak-to-peak ripple was 1.16 rad/s, which was higher than the theoretical estimate. This indicates that the real system includes noise and non-ideal effects that were not fully captured in the simulation. The significant overshoot observed in both implementations, particularly with the lead controller, further demonstrates how sensor noise, actuator limits, and unmodeled dynamics have a significant impact on the hardware response.

Overall, the PI controller provided better steady-state accuracy and a more moderate hardware response, while the lead controller gave a faster response but at the cost of a much larger overshoot in the real system. In simulation, the lead controller appeared superior in both speed and damping, but in implementation, the PI controller behaved more reasonably. This demonstrates that while the PI controller is slower in simulation, but more useful for maintaining controlled speed response on the real SRV02 configuration, the lead controller is more aggressive and sensitive to physical system influences.

References

- [1] Moncayo H. Ph.D. Student Workbook Experiment 03. Embry-Riddle Aeronautical University. 2023.
- [2] Quanser Inc. QUARC User Manual.
- [3] Quanser Inc. SRV02 QUARC Integration, 2008.
- [4] Quanser Inc. QUARC Installation Guide, 2009.
- [5] Quanser Inc. SRV02 User Manual, 2009.
- [6] Quanser Inc. QUARC Compatibility Table, 2010.
- [7] Quanser Inc. SRV02 Rotary Flexible Link User Manual, 2011.

- [8] Norman S. Nise. Control Systems Engineering. John Wiley & Sons, Inc., 2008.
- [9] Quanser Inc. Q2-USB User's Manual, 2010.
- [10] Quanser Inc. VoltPAQ User's Manual, 2010.
- [11] Carl Machover. Basics of Gyroscopes. John F. Rider, 1960.
- [12] Quanser Inc. VoltPAQ User's Manual, 2010. [4] Quanser Inc. QUARC User Manual, 2011.

Appendix A

MATLAB Code:

```
% Author: Anuranan Bharadwaj
% Lab 5

figure()
plot(data_spd(:,1), data_spd(:,2))
hold on
plot(data_spd(:,1), data_spd(:,3))
xlabel('Time (s)')
ylabel('Speed (rad/s)')
title('Step response implementation with Lead Control')
legend('Setpoint', 'Measured', Location='northwest')

%% Noise
time = data_spd(:,1);
r = data_spd(:,2);
y = data_spd(:,3);

index = (time >= 4.9) & (time <= 9.9);

r_ss = mean(r(index));
y_ss = mean(y(index));
e_ss = r_ss - y_ss

%% Bode Plot
K = 1.53;
tau = 0.0254;
s = tf('s');
PI = K/(s*(tau*s+1));
margin(PI)
```